

import lib

```
import pandas as pd
from sklearn.linear_model import LinearRegression
```

load the data

```
data = pd.read_csv("cars.csv")
print(data)
```

clean the data

```
def split_details(data):
    data[["engine", "enginecap", "fuel", "fueltype"]] = data["details"].str.split(" ", expand=True)
    data["enginecap"] = data["enginecap"].str.replace("cc", "").astype(float)
    data = pd.get_dummies(data, columns=["fueltype"], dtype=int)
    if "fueltype_petrol" not in data.columns:
        data["fueltype_petrol"] = 0
    if "fueltype_diesel" not in data.columns:
        data["fueltype_diesel"] = 0
    return data[["enginecap", "fueltype_diesel", "fueltype_petrol"]]
```

features and target

```
features = split_details(data)
print(features)
target = data["price"]
print(target)
```

model

```
model = LinearRegression()
model.fit(features.values, target)
```

prediction

```
cap = int(input("enter engine cap "))
fuel = input("enter fuel type ").lower()
test = pd.DataFrame({"details": ["engine " + str(cap) + "cc" + " fuel " + str(fuel)]})
price = model.predict(split_details(test))
print(price)
```